Hotel Revenue & Occupancy Analysis

Dashboard using Power BI

Industrial Training Project Report

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Year: 2025 – 26



Internship Overview

About the Training Program

Company
Cravita Technologies Private Limited India

Duration
Feb2026 – Sept 2026

Domain
Data Analytics & Data Science

Technologies learned
Excel , PowerBi Till Now

Mode
Offline

CRAVITA TECHNOLOGIES
EXCELLING WITH TECHNOLOGY

Cravita Technologies Private Limited India
(Software company)

- Congress Bhavan, B-1, 2nd Floor, Abhinav Apartment Besides, Congress House Rd, Shivajinagar, Pune, Maharashtra 411005

Cravita Technologies is a results-driven technology and digital solutions company, focused on delivering a complete business ecosystem. We specialize in custom application development—including Education CRMs, Intelligent Chatbots, and HRMS—designed to streamline operations and scale businesses efficiently.

Project Introduction

Hospitality Industry Analytics & Business Intelligence

Hospitality Industry Analytics

The hospitality industry generates massive volumes of booking, revenue, and customer data daily. To remain competitive, hotels must leverage data-driven insights for:

- Monitoring revenue trends across properties
- Understanding occupancy patterns by platform
- Tracking guest satisfaction ratings
- Analyzing booking cancellation patterns
- Benchmarking hotel performance

Why Business Intelligence?

Real-time Insights
Monitor KPIs like revenue, occupancy & ratings instantly

Data-Driven Decisions
Replace guesswork with evidence-based strategy

Performance Tracking
Compare hotels, cities, and booking platforms

Revenue Optimization
Identify revenue leaks and cancellation losses

Goal Alignment
Align operations to achieve occupancy & revenue targets

Problem Definition

Challenges Faced by the Hotel Chain

Revenue Fluctuations

Inconsistent revenue across properties with no visibility into root causes or weekly trends.

Occupancy Challenges

Low and inconsistent occupancy rates,impacting hotel profitability.

High Cancellation Rates

cancellation rate causing revenue losses.

No Centralized Reporting

Data scattered across multiple files and systems — no single dashboard for decision-makers.

Project Objectives

What This Dashboard Aims to Achieve

01

💰 Analyze Revenue

Track total realized revenue vs. generated revenue across all hotel properties and cities.

02

🏨 Monitor Occupancy

Measure occupancy % by property, city, room type, and week to optimize capacity utilization.

03

⭐ Track Customer Ratings

Monitor average guest ratings across hotels and platforms to identify service improvement areas.

04

📱 Analyze Booking Platforms

Evaluate performance across OTAs, direct bookings, and other platforms for strategic partnerships.

05

🚫 Reduce Cancellation Losses

Quantify cancellation-driven revenue loss (LDC) and develop data-backed reduction strategies.

Tools & Technologies Used

Tech Stack for Dashboard Development

Power BI Desktop

Primary tool for data visualization, report creation, and interactive dashboard publishing.

Excel / CSV Files

Source data files: dim_date, dim_hotels, dim_rooms, fact_bookings, fact_aggregated_bookings.

DAX Language

Data Analysis Expressions for creating KPI measures like Revenue, Occupancy %, CR, and LDC.

Data Modeling

Star schema design linking dimension and fact tables via primary/foreign key relationships.

Power Query (M)

Data transformation, cleaning, and shaping before loading into the data model.

Star Schema Design

Dimensional modeling with 2 fact tables and 3 dimension tables for efficient querying.

Dataset Overview

5 Tables | Hotel Booking Domain Data

Table Name	Type	Key Columns	Records / Notes
dim_date	Dimension	date, mmm yy, week no, day_type	~92 rows (May–July 2022)
dim_hotels	Dimension	property_id, property_name, category, city	7 hotel properties
dim_rooms	Dimension	room_id, room_class	4 room types (Elite, Standard, Premium, Presidential)
fact_bookings	Fact	booking_id, property_id, booking_date, check_in, checkout, room_category, booking_platform, ratings_given, booking_status, revenue_generated, revenue_realized	~134K booking records
fact_aggregated_bookings	Fact	property_id, check_in_date, room_category, successful_bookings, capacity	Aggregated occupancy data

Data Model – Star Schema

Relationships Between Tables

Star Schema Design

Central Fact Tables

fact_bookings and fact_aggregated_bookings hold transactional metrics

Dimension Tables

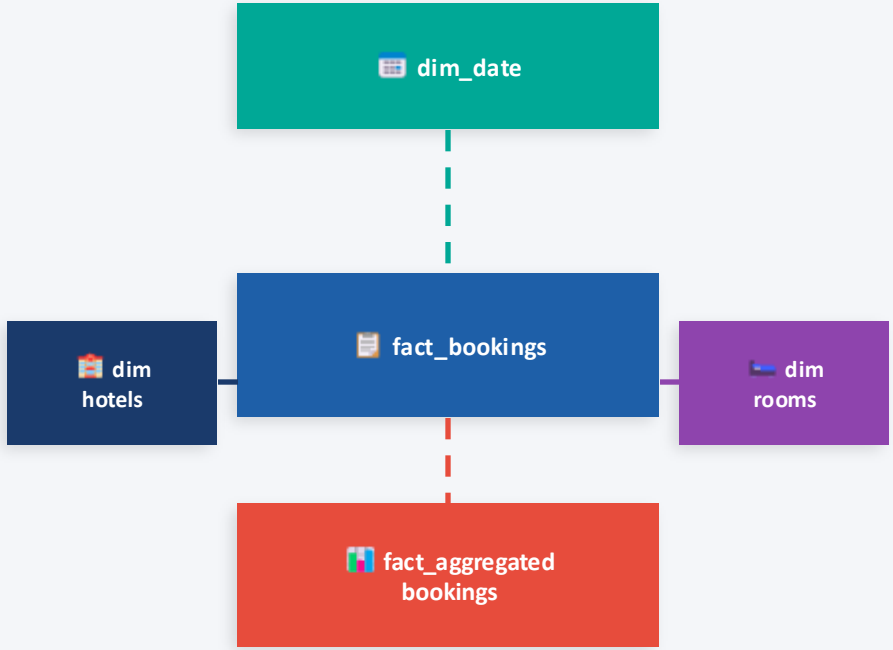
dim_date, dim_hotels, and dim_rooms provide contextual attributes

Relationships

One-to-Many: Each dim record links to many fact records

Key Benefit

Enables fast, flexible slicing of data by city, date, room, platform



All relationships are One-to-Many (1:) from Dimension to Fact tables*

Dashboard Development Process

End-to-End Workflow

1 Data Collection

Gathered 5 CSV files: dim_date, dim_hotels, dim_rooms, fact_bookings, fact_aggregated_bookings

2 Data Cleaning

Removed nulls, standardized date formats, fixed data types using Power Query Editor

3 Data Transformation

Created calculated columns, transformed text fields, prepared data for modeling

4 Data Modeling

Built Star Schema — linked dimension tables to fact tables via property_id, room_id, date

5 DAX Measures

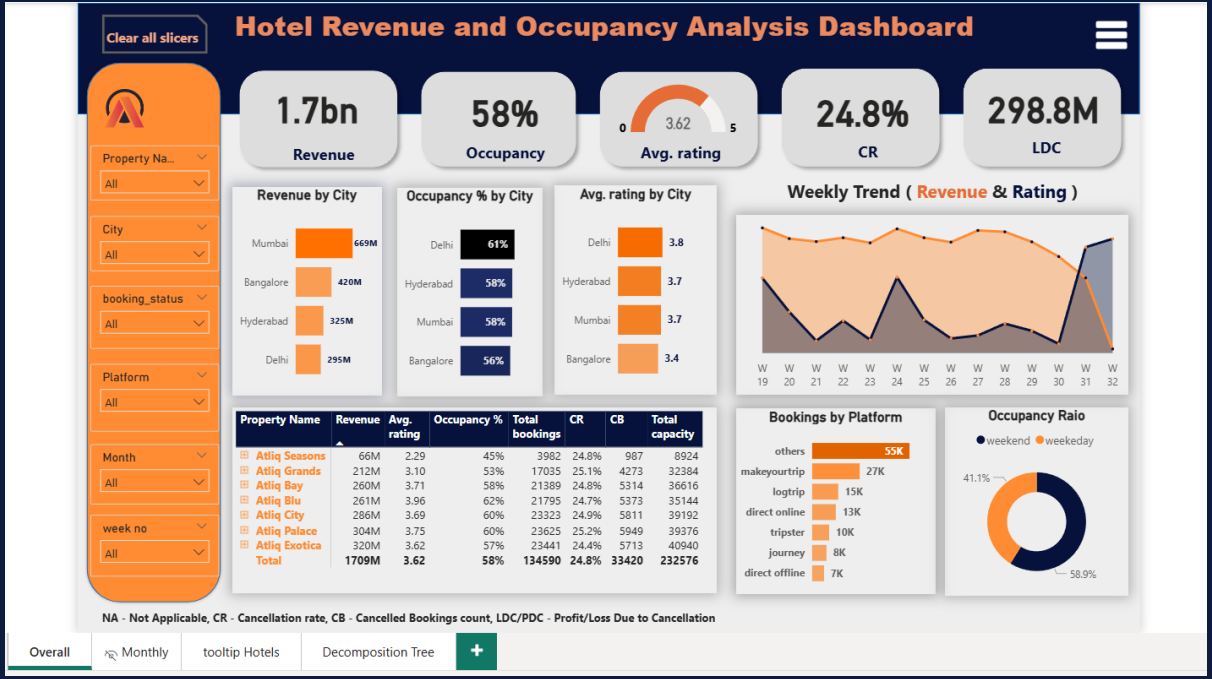
Created 20+ DAX measures: Revenue, Occupancy %, CR, LDC, Avg. Rating, LM comparisons

6 Visualization

Designed 3-page interactive dashboard with KPI cards, charts, tables, slicers & tooltips

Dashboard Overview – Overall Analysis Page

Interactive Power BI Dashboard

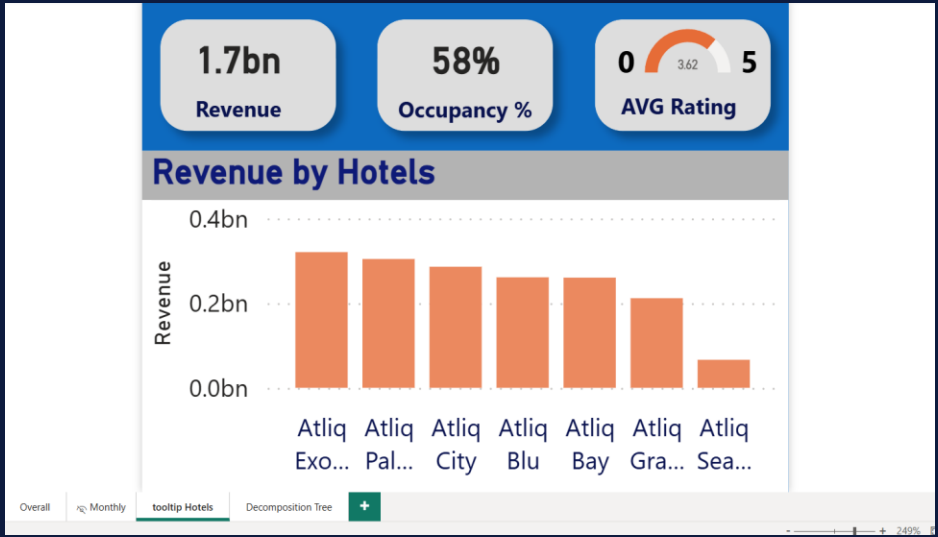


- Revenue by City
- Occupancy % by City
- Average Rating by City
- Weekly Trend (Revenue & Rating)
- Hotel Performance Table
- Bookings by Platform
- Occupancy Ratio
- KPI: 1.Revenue=1.7billion(1B=100cr)
  1.7 Billion = 170 Crore
  2.Occupancy = 58%
  3.AVG Rating = 3.62
  4.CR = 24.8 %
  5.LDC = 298.8M
  2988 Lakhs ÷ 100 = 29.88 Crore

📌 Page 1: Overall Dashboard — Slicers: Property | City | Booking Status | Platform | Month | Week No

Tool Tip – City wise Hotels performance analysis

Interactive Power BI Dashboard



Developed a custom Power BI tooltip to enable city-wise hotel performance analysis, allowing users to view revenue, occupancy, and rating metrics for hotels within a selected city through an interactive hover feature.

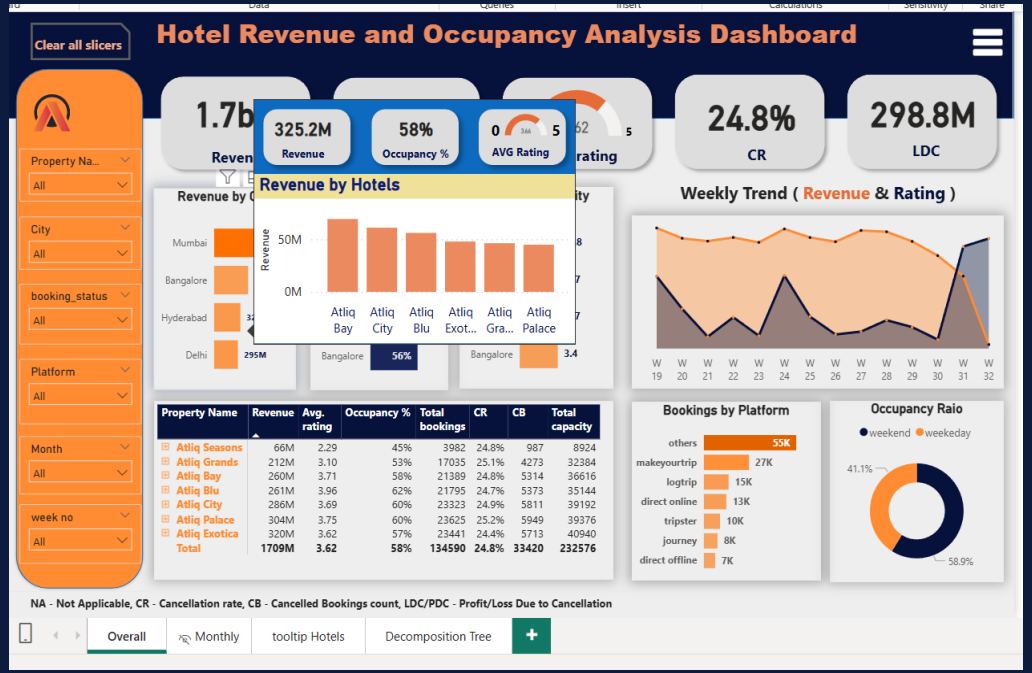
Metrics Displayed in Tooltip

- Revenue
- Occupancy %
- Average Rating
- Revenue by Hotels (column chart)

📌 Tooltip — Revenue by hotels in the city

Dashboard Overview – Overall Analysis Page

Interactive Power BI Dashboard



As Shown in figure the cursor is on hydrabad city Then tooltip shows:

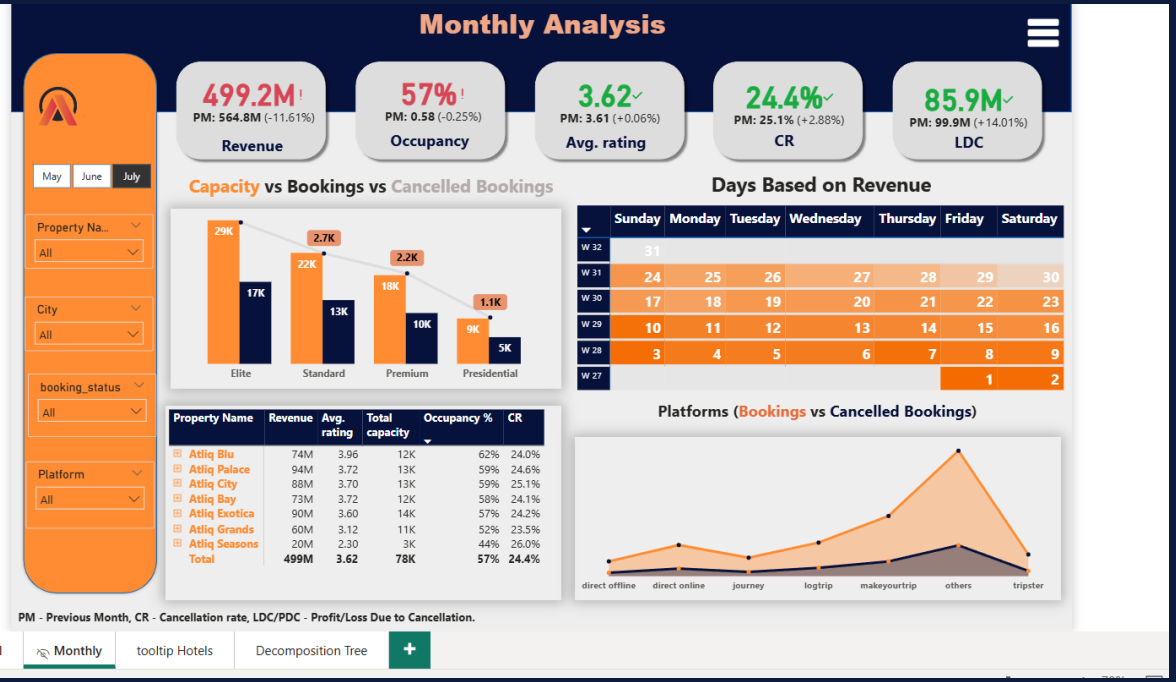
- Total Revenue : 325.2 M
- Occupancy : 58%
- Avg Rating : 3.66
- Column Chart: comparison Between Hotels in Hydrabad

It works like these for all other Cities

📌 Page 1: Overall Dashboard — Slicers: Property | City | Booking Status | Platform | Month | Week No

Dashboard – Monthly Analysis Page

Month-wise Drill-down with Previous Month Comparison

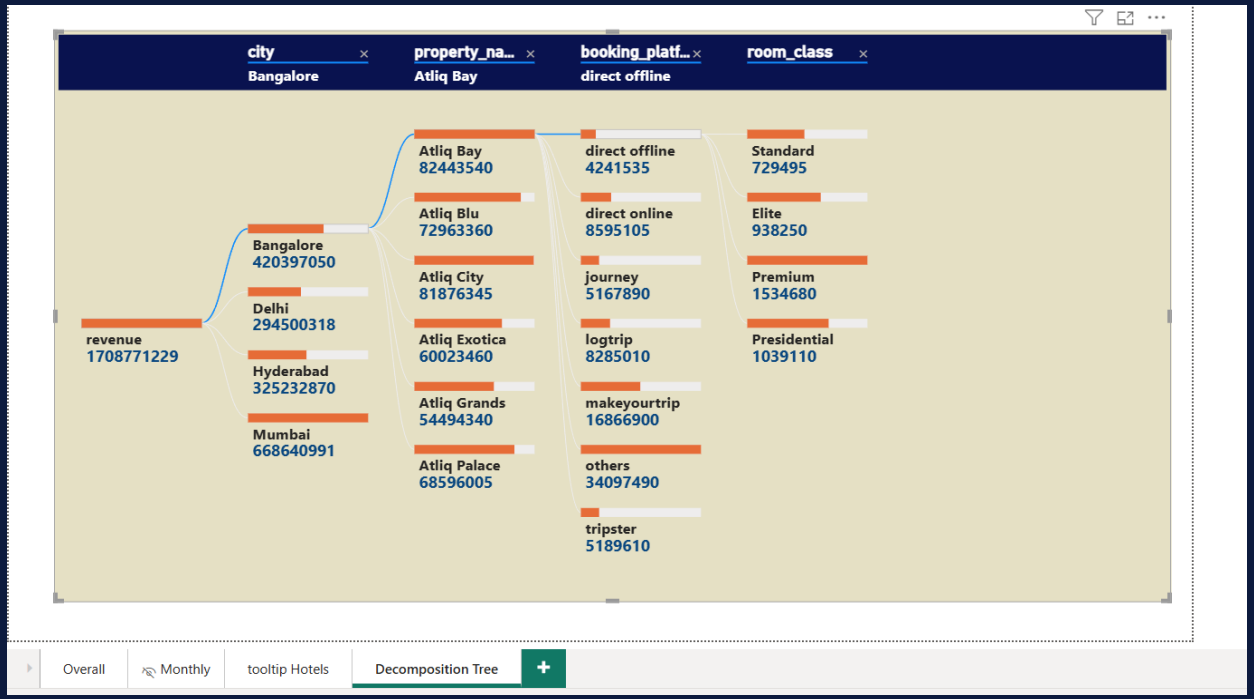


- This page focuses on monthly performance analysis.
- Users can compare revenue, occupancy, customer ratings, and cancellation rates for different months.
- The dashboard also analyzes room category performance, daily revenue patterns, hotel-wise metrics, and booking platform effectiveness.
- This helps management understand monthly trends, identify peak demand periods, and make informed decisions regarding pricing, capacity planning, and customer retention.

📌 Page 2: Monthly View — Shows PM comparison, Capacity vs Bookings, Days Based on Revenue Calendar, Platform trends

Decomposition Tree

Total Revenue –> Cities – >Property Name –> Booking Platform –>Room Class



Start :Total Revenue

Level 1:Revenue by City (click on city)

Level 2: Revenue by Property (Properties in clicked city

Level 3:Revenue by Booking Platform(click on any hotel)

Level 4:Revenue by Room Class(click on any platform)

Page 2: Monthly View —Decomposition Tree ->total Revenue –> Cities – >Property Name –> Booking Platform –>Room Class

Dashboard KPIs & Key Metrics

Core Business Performance Indicators

₹1.7 Bn	58%	3.62	24.8%	₹298.8M
Total Revenue Realized	Avg. Occupancy Rate	Avg. Customer Rating (/ 5)	Cancellation Rate (CR)	Loss Due to Cancellation (LDC)

KPI Definitions & Significance

- Revenue: Sum of revenue_realized from fact_bookings. Excludes cancellation-lost revenue.
- Occupancy %: Successful bookings ÷ Total capacity. Measures how well rooms are utilized.
- Avg. Rating: Average of ratings_given. Max 5 — benchmarks guest satisfaction.
- Cancellation Rate (CR): Cancelled bookings ÷ Total bookings. Currently 24.8% — critically high.
- LDC (Loss Due to Cancellation): revenue_generated – revenue_realized. ₹298.8M potential revenue lost.

Key Analysis & Findings

Insights Derived from the Dashboard

Mumbai Generates Highest Revenue

₹669M

Mumbai Revenue

Mumbai leads all cities in total realized revenue at ₹669M, driven by higher room rates and occupancy. Delhi follows at ₹295M.

Delhi Has Highest Occupancy

61%

Delhi Occupancy

Delhi achieves 61% occupancy rate — the highest among all cities. Bangalore and Mumbai both stand at 58%.

Bangalore Has Lowest Rating

3.4

Bangalore Rating

Bangalore records the lowest average customer rating of 3.4 out of 5. This signals service quality issues requiring urgent attention.

Weekend vs. Weekday Occupancy

58.9%

Weekend Occupancy

Weekend occupancy (58.9%) is significantly higher than weekday occupancy (41.1%). Weekday demand generation is a key opportunity.

Atliq Exotica — Top Performer

₹320M

Exotica Revenue

Atliq Exotica is the top revenue-generating property with ₹320M revenue, 57% occupancy, and 3.62 avg rating.

Atliq Seasons — Lowest Performance

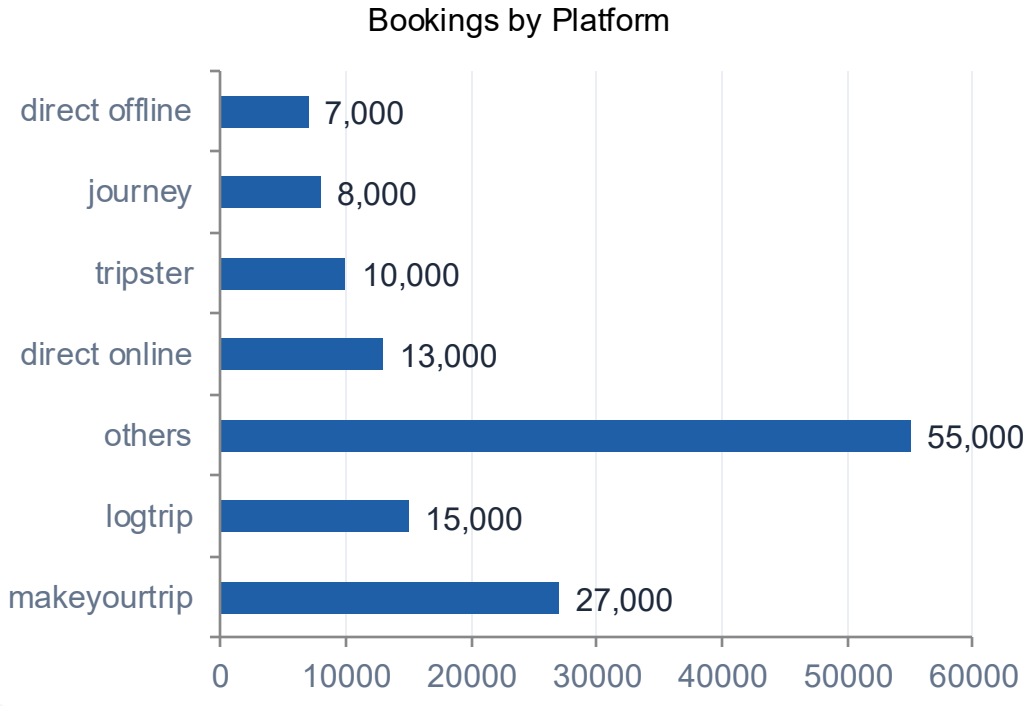
₹66M

Seasons Revenue

Atliq Seasons has the lowest revenue (₹66M) and a notably low 2.29 average rating — indicating both demand and satisfaction issues.

Booking Platform Analysis

Revenue & Booking Distribution Across Channels



Platform Insights

others

Highest bookings (~55K) — likely OTA aggregators

makeyourtrip

Second largest platform with ~27K bookings

logtrip

Third with ~15K — significant online channel

direct offline

Lowest bookings (7K) — traditional walk-ins

Weekend vs Weekday

Weekends show 58.9% vs 41.1% weekday occupancy — a 17.8% gap

DAX Measures – Custom KPI Calculations

Power BI Data Analysis Expressions

Measure Name	DAX Formula	Business Purpose
Revenue	`SUM(fact_bookings[revenue_realized])`	Total realized revenue from all bookings
Occupancy %	`DIVIDE([Total successful bookings], [Total capacity], 0)`	Room utilization rate
Cancellation Rate	`DIVIDE([Total cancelled bookings], [Total bookings], 0)`	Proportion of bookings cancelled
Avg. Rating	`AVERAGE(fact_bookings[ratings_given])`	Mean customer satisfaction score
Revenue LM	`CALCULATE([revenue], DATEADD(dim_date[date], -1, MONTH))`	Previous month revenue for comparison
LDC	`[revenue BR] - [revenue]`	Loss due to cancellation (generated - realized)

📌 *DATEADD and CALCULATE functions enable time-intelligence comparisons (current vs previous month) critical for trend analysis*

Business Recommendations / Findings

Data-Driven Action Plan for Management

01

Reduce Cancellations

- Implement advance payment / non-refundable booking options
- Send automated reminders 48hrs before check-in
- Target the 24.8% CR to recover ₹298.8M in lost revenue

02

Boost Weekday Occupancy

- Offer corporate packages and business traveler deals
- Launch weekday-only discount promotions
- Bridge the 17.8% gap between weekend (58.9%) and weekday (41.1%) occupancy

03

Improve Low-Rated Cities

- Investigate service quality issues in Bangalore (3.4 rating)
- Staff training and guest feedback programs
- Target Atliq Seasons (2.29 rating) for urgent service overhaul

04

Strengthen Online Partnerships

- Invest in top-performing platforms (MakeYourTrip, Logtrip)
- Boost Your Own Website Performace
- Give offers to Customers on your website to avoid commision of other websites.

Challenges Faced

Technical & Design Hurdles During Development

Complex Data Modeling

Challenge:

Establishing correct relationships between 5 tables, handling ambiguous join paths, and managing filter propagation direction in the Star Schema.

Solution:

Google , Youtube to learn about Star Schema

DAX Measure Complexity

Challenge:

Writing time-intelligence functions (DATEADD, CALCULATE) for last-month comparisons that work correctly across all slicer combinations.

Solution:

Used ChatGPT ,Google, Claude ai to help in operation

Dashboard Design Optimization

Challenge:

Fitting 5 KPIs, 6+ charts, 2 tables, and 6 slicers across 3 pages while maintaining clarity and avoiding visual clutter.

Solution:

Used tooltip pages, bookmarks, and dynamic titles to pack information efficiently without overcrowding visuals.

Date Table Configuration

Challenge:

Configuring the dim_date table as a proper date table and ensuring week number calculations matched the hotel's fiscal calendar.

Solution:

Marked dim_date as 'Date Table' in Power BI settings and validated week numbers against source data.

Future Enhancements

Roadmap for Advanced Analytics Capabilities

Real-Time Data Integration

Phase 1

Connect Power BI to live hotel management systems via DirectQuery or streaming datasets for real-time occupancy and booking tracking.

ML-Based Demand Forecasting

Phase 2

Integrate Azure ML or Python models to forecast occupancy rates and revenue for next 30/60/90 days using historical booking patterns.

Dynamic Pricing Analytics

Phase 2

Build a pricing recommendation engine that suggests optimal room rates based on occupancy trends, seasonal patterns, and competitor data.

Customer Segmentation

Phase 3

Cluster guests by booking behavior, spend, platform, and loyalty for targeted marketing and personalized offers to high-value segments.

Automated Alert System

Phase 1

Set up Power BI data alerts to notify managers when occupancy drops below threshold or cancellation rate exceeds acceptable levels.

Mobile Dashboard App

Phase 1

Optimize dashboard for Power BI Mobile app so hotel managers can monitor KPIs and respond to trends from anywhere on iOS or Android.

Conclusion

Project Summary & Business Impact

🏆 Project Achievements

- Built a 3-page interactive Power BI dashboard
- Analyzed ₹1.7 Billion in hotel revenue data
- Uncovered ₹298.8M in cancellation losses
- Identified 17.8% weekday occupancy gap
- Compared 7 hotel properties across 4 cities
- Enabled drill-down by platform, city, and week

💼 Business Impact

Revenue Visibility

Managers can now monitor revenue trends in real-time instead of relying on monthly Excel reports.

Cancellation Action

Dashboards quantify ₹298.8M LDC — creating urgency for implementing advance payment policies.

City-level Strategy

City and property-wise breakdown enables targeted performance improvement plans.

Platform Intelligence

Platform analysis reveals where to invest marketing budget for maximum booking ROI.

Thank You!



Give your Valuable Feedback

Project: Hotel Revenue & Occupancy Analysis Dashboard

Student: Jay Satish Gunjal

Tool: Microsoft Power BI

Academic Year 2025 – 26 | Data Analytics Internship Project

Dashboard built using Power BI Desktop | Data: 5 CSV files | ~134K booking records | 3 months (May–July 2022)

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